Dependence Implication

Conceptual map: solved and open questions

Definitions

Team. A set $T \subseteq D^V$ of assignments $s : V \to D$.

Dependence atom. $T \models =(X; y)$ means that rows agreeing on X also agree on y.

Disjunction. $T \models \bigvee_{i=1}^{k} \alpha_i$ if $T = \bigcup_{i=1}^{k} T_i$ with $T_i \models \alpha_i$.

Implication. $\Sigma \models \varphi$ if every team satisfying all premises in Σ also satisfies φ.

D is any nonempty value domain; V contains the finitely many input variables. Disjunctions are nonempty. Cover parts may overlap or be empty; repeated atoms count separately.

Finite and unrestricted implication coincide $\Sigma \models_{\text{fin}} \varphi \iff \Sigma \models \varphi$

Every failure of implication has a finite counterteam.

Q1 General complexity — SOLVED

co-r.e.-complete under computable many-one reductions; hence undecidable.

Hardness already holds for the fixed consequent

$$\varphi_* = =(w; z) \vee =(w; z) \vee =(U; w) \vee =(U; w) \vee =(V; w) \vee =(V; w).$$

At most **five atom occurrences per premise**; every variable except U, V, X, Y, w has an explicit bound of at most three values.

Counterteam size. No computable input-dependent bound exists on the size of a smallest finite counterteam.

Q1.1 Decidability — OPEN

Which syntactic restrictions on the premises and consequent guarantee **decidable implication**?

Q1.2 Tractability — OPEN

Which syntactic restrictions on the premises and consequent guarantee **polynomial-time implication**?

Time is measured in the total explicit input length.